

Genset ECU — Reconciled Specification

Date: 2026-09-13 **Status:** Phase 0 — supersedes all four prior artifacts
Target: Functional replacement for the `ENGINE CONTROL UNIT` block of Kirloskar wiring diagram `08-GP3-60-001` , on a 25 kVA diesel genset with a 12 V / 75 Ah battery.

1. Why this document exists

Four artifacts precede this one:

Artifact	Rev	Role
Genset ECU Design Brief	J	Spec
ECU Circuit Guide	H	Per-block why/how/prove
ECU Circuit Schematics	H	15 topology sheets
PCB Design Roadmap	A	Brief → fab sequence

They were written by reverse-engineering the Kirloskar wiring diagram from memory of it, not from the document itself. Now that `GP3.8703.C4.pdf` is in hand, roughly a third of their content is wrong — not stylistically, but in ways that change the BOM, the driver topology and the pin count.

This document is the single source of truth from here. Where it contradicts an artifact, this document wins. The artifacts remain useful for the reasoning behind blocks that survived unchanged, and the Circuit Guide's "prove it" bench procedures remain valid for those blocks.

Raw extracted evidence: `refs/ecu-pinout-extracted.md` .

2. What changed, and why it matters

2.1 Injector drive is bank-shared

The diagram traces injector high-side pin `03` to both cylinder 1 and cylinder 3, joined at splice V5. Pin `05` serves cylinder 2 alone. Low-side selects are individual: `73` (cyl 1), `07` (cyl 3), `29` (cyl 2).

Five pins, two banks. The artifacts specify an independent high/low pair per cylinder — six pins, three banks — which would put three boost stages on a board that needs two.

This is the conventional arrangement: the boost converter and high-side switch are the expensive, thermally-significant parts, so they are shared across cylinders that never fire simultaneously. Cylinder selection happens on the low side, which is cheap.

Consequence: two high-side bank switches, one shared boost rail, three low-side current-controlled selects.

2.2 Fuel metering unit is low-side PWM, not an H-bridge

Metering unit pin 2 sits on fused battery (20 A, 8F1). Pin 1 goes to ECU pin 88 on a black wire — the low-side class per the page-2 legend. The ECU sinks current; it does not source it, and it never reverses polarity.

The Design Brief specifies a TLE9201SG H-bridge with one leg tied off. That works electrically but buys a reversing bridge for a solenoid that only pulls one way.

Consequence: one high-current low-side driver with current sense, closed-loop in firmware. One fewer part family to source.

2.3 Fuel level and water-in-fuel are not ECU signals

Both route through the XC11/XC1 36-pin harness to the panel — fuel level to XC pin 22 (wire 1C2), water-in-fuel to XC pin 16/1 (wire 4B). Neither reaches an ECU pin anywhere in the diagram. They belong to the GCU.

Consequence: two analog front-ends and two connector positions removed.

2.4 Discrete inputs are active-high, not active-low

AUDIO ABORT (20), OVERSIDE SWITCH (24) and IGNITION (71) are all red wires — switched battery. The switch connects the pin *to battery*, not to ground.

The artifacts specify pull-up to 5 V with the switch closing to ground. That is the opposite sense, and the front-end is wrong for it: these pins see battery voltage, not 5 V, so they need a divider rated for the full 6–40 V input range plus a clamp.

The fail-safe reasoning also inverts. Circuit Guide Group D argues that normally-closed wiring makes an open harness read as a fault. With switched-battery inputs an open wire reads "not asserted" — indistinguishable from a switch that simply isn't pressed. That argument does not hold here and should not be repeated in the new design. Detecting an open wire on these inputs requires either a pull-down and a diagnostic current source, or accepting that it is not detectable. **We accept it is not detectable** and note it in the FMEA, because changing the sense would break harness compatibility.

Pins 01 (G_G_B AT1) and 02 (G_G_B AT2) are ECU-provided return/ground legs for the switch circuits. The coolant switch (43) and droop switch (23) are cyan — digital-input class — and are conventional switch-to-return inputs.

2.5 Both CAN channels are real

Pins 55 / 77 and 87 / 86 are both drawn as twisted pairs with bidirectional markers. The artifacts treat CAN1 as a spare, DNP on the harness. In the OEM system both are wired.

Consequence: populate both transceivers and both connector positions.

2.6 Scope reductions confirmed

None of the following exist anywhere in this system: magnetic pickup unit, DEF/urea sensor, LIN bus, SCR, separate overspeed switch, barometric sensor in the harness. There is **one** catalyst temperature sensor connector (2-pin), not three EGT channels.

OVERSIDE SWITCH (pin 24) and Override SS SW -S7 are one component. The Design Brief lists them as two.

3. Safety position

Revised 2026-09-13 after research memo 02 and a follow-up trace of the OEM diagram. This section originally claimed engine speed reaches the GCU over CAN only, and that the architecture therefore has no independent overspeed path. **That was wrong, and wrong in the pessimistic direction.** The corrected picture is below; the gate it imposes is relaxed but not removed.

There is no magnetic pickup. Confirmed as far as desk research can: the OEM diagram labels every terminal in the KG640C's J1-J8 with no speed-pickup among them, and KOEL's manual for the sibling KG640 (SED-MAN-KG640-002) enumerates all 42 terminals with no MPU input.

But the GCU has an independent overspeed path anyway, and it is better than an MPU trip in one respect and worse in another. Three findings combine:

1. **The GCU senses speed from alternator frequency, not from CAN.** The KG640 manual's sole Engine Speed Sense Source value is Alternator frequency ; terminals 27-30 sample the genset's own AC output at 5 kHz over 3-75 Hz. On a 4-pole alternator with no slip, 1500 rpm is rigidly 50 Hz. This path does not touch the engine ECU or the CAN bus.

2. **The GCU can cut fuel in hardware.** Its digital output B drives the ignition relay -13RB1, which is rated 70 A — a power contactor, not a signal relay. Dropping it removes the supply feeding the ECU and the fuel metering unit rather than merely signalling the ECU to stop.
3. **So the whole chain bypasses this ECU.** Alternator frequency → GCU AC front-end → GCU firmware → 70 A contactor → fuel system unpowered. A hung ECU or a dead CAN bus does not defeat any link in it.

What this leaves. The residual single point of failure is the GCU's own microcontroller. A GCU firmware hang defeats the path; an ECU hang does not. That is a materially different and much smaller risk than the one this section originally described.

The revised gate. A standalone MPU-driven overspeed trip module is **recommended, not mandatory**, and the decision is now a judgement about GCU reliability rather than about a missing path. Two candidates are costed in memo 02 (GAC SSW675, Murphy HD9063). What remains **mandatory** before any build takes injection authority is confirming finding 2 by measurement — see U12 — because the whole relaxation rests on the ignition relay actually removing power from the fuel system rather than merely signalling.

The Circuit Guide argued — correctly, as a matter of engineering — that overspeed protection should not depend on the same MCU and the same bus that could be the thing that failed. The OEM system does not honour that. We are replacing the OEM ECU, so we inherit its interface, and adding an MPU requires a GCU input we have not confirmed exists.

The position for this design:

1. The ECU implements overspeed shutdown in firmware as a **primary** function, with the crank sensor as its input, at the highest scheduling priority. This is the first line, and it is ours.
2. The GCU's alternator-frequency path plus its 70 A ignition contactor is the **second line**, independent of this ECU. We must not do anything that weakens it — in particular, the ECU must not latch the fuel metering unit on in a way that survives its supply being removed, and must fail safe when the ignition input at pin 71 drops.
3. A standalone MPU trip module is the **optional third line**, covering GCU firmware failure. Recommended; a judgement call for the project owners on cost against residual risk, no longer a blocking requirement.
4. **The gate that remains: U12 must be closed by measurement before any build takes injection authority.** If the ignition relay turns out to signal the ECU rather than remove power from the fuel system, the second line does not exist, item 3 reverts to mandatory, and this section reverts with it.

A 25 kVA set in runaway is a mechanical hazard to anyone near it. The failure mode that matters is a controller hanging with fuel still flowing. What makes this architecture acceptable is that stopping the fuel does not require the hung controller's cooperation — which is exactly the property U12 verifies.

4. ECU pin map

Confirmed pins from the wiring diagram. 44 of 94. See §8 for the remaining 50.

Power

Pin	Signal	Front-end
21	Battery + feed	Main input: EMI filter → fuse → TVS → reverse-battery FET
04	V_V_BAT_1R (10 A fused)	120 k / 10 k divider → ADC; feeds high-current loads [?]
06	V_V_BAT_2R (10 A fused)	120 k / 10 k divider → ADC; feeds high-current loads [?]
09	SYNCHRONIZATION GROUND	Function unresolved — see §8

The battery-sense divider is 120 k / 10 k, not the 75 k / 10 k inherited from the earlier artifacts. That value was sized for a 5 V ADC and delivers 4.69 V at a 40 V input — above the S32K148's 3.3 V analog supply. The corrected divider peaks at 3.07 V and still resolves 573 ADC counts at the 6 V cranking dip. Verified in `sim/blocks/battery_sense.cir`.

Ratiometric analog — 10 k / 16 k divider, 22 nF shunt, 3.3 V clamp, rail measured and corrected in firmware

Revised 14 Sep 2026. This section previously specified *1 kΩ series, 100 nF shunt, ADC ref tracks 5V_SENSOR*, which contradicted the simulated front-end and cannot be built on this MCU. Both halves were wrong, and for the same reason the 75 k / 10 k battery divider was wrong — they were written for a 5 V part:

- **VREFH cannot track 5V_SENSOR**. The S32K148's ADC reference may not exceed its 3.3 V analog supply. A reference at 5 V is not an option on any 3.3 V device.
- **1 kΩ in series scales nothing**. With no division, a sensor at 4.5 V arrives at a 3.3 V pin. The series resistor is a fault-current limiter, not an attenuator.

So the signal must be divided, and the question becomes how to keep the *ratiometric* property after dividing it.

Decision: divide, then measure the rail and correct in firmware.

`5V_SENSOR` gets its own ADC channel. Every ratiometric reading is computed as a fraction of the measured rail rather than of an assumed

5.000 V, which is what the sensor's output actually represents. Cost: one ADC channel, zero analog parts. The channel budget has room — see the sensor reference.

Rejected alternative: dividing `5V_SENSOR` into `VREFH` by the same 16/26 ratio. It fits numerically ($5\text{ V} \times 16/26 = 3.08\text{ V}$, inside the 3.3 V limit) and gives true ratiometric conversion with no firmware arithmetic. It is rejected because `VREFH` is shared by every channel on that ADC, so it would silently make **battery sense and the NTC channels ratiometric to the sensor rail as well** — both of which need an absolute reference. It would also need a buffer, since a resistive divider is too noisy and too high-impedance to drive a reference input.

Single-ended or differential is decided by the ground, not by the channel. Where the sensor's return is shared, the other sensor's current develops an offset across the harness resistance that a single-ended input reads as signal: `sim/blocks/sensor_differential.cir` measures 12 mV of bias at 1 Ω of return resistance against 0.13 mV differential. Research memo 09 found both Deep Sea Electronics and SEDEMAC treat this as a first-class problem. So:

- **Shared ground (pins 34, 36) → differential front-end.**
- **Dedicated ground (pin 08) → single-ended is sufficient**, because there is no other sensor's current in that return. This mirrors the OEM's own ranking: they gave rail pressure a dedicated return precisely so it would not need one.

Pin	Signal	Group ground	Front-end
11 / 41	Boost: 5 V excitation / pressure signal	34	Differential
79	Boost: temperature signal — topology unresolved, see §8	34	Differential
34	Sensor ground — boost + coolant	—	—
33	Coolant temperature (NTC, pull-up divider)	34 (spliced)	Differential
15 / 37	EGR position: 5 V / wiper	36	Differential
36	Sensor ground — EGR + oil pressure	—	—
39 / 80	Oil pressure: 5 V / signal	36 (spliced)	Differential
32 / 35	Rail pressure: 5 V / signal	08	Single-ended
08	Sensor ground — rail pressure, dedicated	—	—
—	<code>5V_SENSOR</code> rail monitor	—	Divider → ADC, for the correction above

Sensor-ground grouping mirrors the OEM harness exactly: 34 boost+coolant, 36 EGR+oil, 08 rail alone, 30 crank alone, 44 cam alone. Rail pressure gets a dedicated return because it is the most accuracy-critical analog channel on the engine.

Speed

Pin	Signal	Front-end
52 / 74	Crank VR: Frequency I/P High / Low	Differential adaptive-threshold conditioner → timer capture
30	Crank sensor ground	Shield terminates here, ECU end only
45 / 46	Cam Hall: 5 V / Frequency I/P	Pull-up to 5V_SENSOR → timer capture
44	Cam sensor ground	—

Discrete inputs

Pin	Signal	Sense	Front-end
43	Coolant switch	Switch-to-return	Pull-up + RC debounce
23	Droop switch	Switch-to-return	Pull-up + RC debounce
01 / 02	G_G_B AT1 / AT2	—	Switch return legs — but see note below, these may be power grounds
20	Audio abort	Active-high	47 k / 68 k bias + 3.0 V zener clamp + 220 nF
24	Override SS	Active-high	same
71	Ignition	Active-high	same

The active-high front-end is not a divider, and cannot be. Reading logic high at a 6 V cranking dip needs a divider ratio above $2.31/6 = 0.385$; staying under 3.3 V at 40 V needs it below $3.3/40 = 0.0825$. Those constraints do not overlap, so no fixed ratio works at both ends. The zener sets the clamp level and the divider only biases into it, which makes the logic level flat to within 92 mV across the whole range instead of tracking the battery. Verified in [sim/blocks/discrete_input.cir](#).

Pins 01/02 need checking. Memo 03 infers from the connector drawing that pins 1, 2, 5 and 6 are large power contacts. Pins 05 and 06 fit that — injector high side and a battery rail. Switch return legs would not. The likelier reading is that G_G_B denotes battery ground and these are the ECU's main power grounds, which would also explain why no power-ground pin appears anywhere in the extraction. Queued as a continuity measurement.

Outputs

Pin	Signal	Driver
03	Injector high side — bank A (cyl 1 + 3)	Boosted high-side switch

Pin	Signal	Driver
05	Injector high side — bank B (cyl 2)	Boosted high-side switch
73 / 07 / 29	Injector low side — cyl 1 / 3 / 2	Current-controlled low-side, peak-and-hold
88	Fuel metering unit PWM	High-current low-side + current sense
59 / 81	EGR High / EGR Low	H-bridge + position feedback on pin 37
50	Main relay	Low-side FET + flyback
69	Buzzer relay	Low-side FET + flyback

CAN

Pin	Signal
55 / 77	CAN channel 1 — H / L
87 / 86	CAN channel 2 — H / L

5. MCU selection

Recommendation: NXP S32K148. This resolves the contradiction in the artifacts — Design Brief §03 specifies MPC5744P while its own §02 block diagram and §06 sensor table still say S32K148 and FTM.

The argument for MPC5744P was angle-synchronous injection timing needing the eTPU2.

Corrected 2026-09-13 after research memo 05 challenged the original reasoning and was right to. This section first argued that constant 1500 rpm operation relaxes the timing problem because a crank degree is 111 μ s and a 60-tooth wheel gives an edge every 667 μ s. That checks the wrong variable. Crank-edge *capture* rate was never the binding constraint in either case — at 6000 rpm a 60-tooth wheel still only produces an edge every 28 μ s, which is over 2000 CPU cycles at 80 MHz against an NVIC latency of roughly 12. The comparison was never close, so constant speed cannot be what makes it comfortable. The reasoning below replaces it.

What the eTPU2 actually buys. It is a co-processor that autonomously schedules and re-schedules many angle-domain output-compare events across a wide and *rapidly changing* speed range, without consuming the main CPU's interrupt budget and without the core's scheduling jitter — cache effects, higher-priority ISRs, RTOS latency — reaching the injection timing chain. That matters when rpm sweeps 800–6000 with tight transient response and pilot/main/post multi-pulse strategies all landing within microseconds of a moving target angle.

Why we don't need it. This set governs to a fixed 1500 rpm, and the pin map in §4 shows a single main injection pulse per cylinder per cycle — one bank high-side plus one low-side select, with no multi-pulse rate shaping in scope. The constraint that does bind is **injector pulse-width resolution**, which sets fuel-quantity metering accuracy and is independent of crank speed. FlexTimer output-compare runs off the bus clock: one tick is 12.5 ns at 80 MHz, 8.9 ns at 112 MHz HSRUN. Against a single-pulse metering requirement on the order of 1–4 µs, that is roughly 100–450× headroom.

The S32K148 also carries 2× Programmable Delay Blocks, which trigger ADC conversions or timer events at a defined delay from a capture event in hardware. That recovers part of what the eTPU2 offers — jitter isolation on the timing chain — without the eTPU2.

Caveat carried forward: the 1–4 µs resolution figure is a general diesel-injector engineering figure, not this injector's datasheet. Revisit if U5 closes with a part demanding materially finer control, or if multi-pulse injection ever enters scope.

What the S32K148 buys in exchange:

- Free S32 Design Studio; GCC ARM toolchain; cheap CMSIS-DAP/J-Link probes instead of a Power Architecture Nexus pod
- Stock KiCad symbol availability, or trivially built from NXP's own EVB files
- Real availability in small quantities in India
- 3× FlexCAN (we need 2); dual 12-bit ADC against a demand of ~14 channels — 8 sensor inputs, 2 battery-rail senses, and 4 current-sense returns from the injector, metering and EGR drivers; FlexTimer instances well beyond the ~10 timing channels required (crank capture, cam capture, 3 injector low-side, 2 injector high-side, metering PWM, 2 EGR bridge PWM)
- No eTPU2 microcode, which is a specialist skill and a hiring risk on a small team

Confirmed in Phase 1 — see <docs/research/05-mcu-selection.md>.

Orderable part **FS32K148HAT0MLQT**, 144-pin LQFP, 80 MHz RUN / 112 MHz HSRUN grade, –40...+125 °C. In stock at DigiKey India at roughly ₹1,900–2,000/unit. The S32K146 fallback was checked and is in worse supply shape (backordered, 12–52 week spread), so it is not needed.

Two items deferred to Phase 2: confirming how many ADC channels the 144-pin package actually breaks out (needs the reference manual's pin-mux table, not the datasheet), and pad-by-pad verification of the SOT486-2 package drawing before the footprint is locked.

6. Power architecture

Unchanged from the artifacts in topology; the reasoning there is sound.

Stage	Function
EMI input filter	Common-mode choke, CISPR 25 conducted margin
Reverse-battery	Ideal-diode controller + P-FET, < 0.3 V drop
Transient clamp	Bidirectional TVS, ISO 7637-2 pulses incl. load dump 5b
Buck pre-regulator	6–40 V → 5 V, wide-input synchronous
5V_SENSOR	PTC-protected per sensor group, isolated from MCU rails
3V3_MCU	LDO off 5 V
Supervisor	Windowed watchdog + brown-out reset

Input range 6–40 V: 6 V covers the cranking dip on a 12 V system, 40 V bounds the clamped load-dump transient.

Injector boost rail is new relative to the power chain as drawn: a separate boost stage generating the injector opening voltage, with its own reservoir bank sized for the pulsed draw of two banks. Target voltage is set by the injector part — see §8.

7. Deliberate deviations from the OEM design

Recorded so they are choices, not drift:

1. **Connector split.** The OEM uses one 94-pin connector. We use functionally-split sealed connectors. This simplifies harness build and field diagnosis, at the cost of needing an adapter harness to mate with an unmodified Kirloskar loom. That adapter is a deliverable, not an afterthought.
2. **Onboard barometric sensor.** Not present in the OEM harness. Added on-PCB behind a vented port; costs nothing in harness terms and improves fuelling correction.
3. **Independent overspeed path.** Added, per §3. The OEM does not have one.

8. Unknowns register

Nothing in this list may be guessed into copper. Each needs an answer before the sheet that depends on it is drawn.

Status updated at phase 1 exit, 14 Sep 2026. Three states, and the middle one carries weight: **Open** means no answer yet; **Answered** means phase-1 research produced an answer that is good enough to design

against but is *inferred*, not confirmed, and still carries the upgrade path in its memo; **Closed** means confirmed outright. Nothing here is Closed yet — every confirmation on this list needs the machine.

#	Unknown	Status	Blocks	How to close
U1	Remaining 50 of 94 ECU pins	Open	Connector selection, adapter harness	ECU connector part number — photograph the housing on the engine. Memo 03 ranks a shortlist; only a logo or molded part number settles it
U2	Boost air-temp (pin 79): NTC or ratiometric?	Open	Sheet 3b front-end	One resistance reading, sensor cold — brief task 1.3, closes it outright. Mitigation: a front-end that works either way
U3	KG640C MPU input — does it exist?	Answered — no MPU input, inferred, high confidence (memo 02)	§3 safety gate	Upgrade to confirmed: a manual titled KG640C (not the base KG640), a photograph of the physical rear terminal block, or KOEL technical support
U4	KG640C J1939 PGN set and expectations	Open	Firmware, CAN sheet	KG640C manual; or log the live bus with the OEM ECU fitted
U5	Injector part number and drive profile	Partly answered — memo 04 gives a family envelope (65–115 V, 12–24 A peak, 8–13 A hold, 40–100 A/ms) sufficient to design the driver	Boost rail <i>target</i> voltage, not the driver sheet itself	Photograph the injector body for a Bosch 0445 1xx xxx or Kirloskar F6.xxx.xx.x.pr . Separately: the calibration data is a different problem — see §9 and memo 04
U6	Pin 09 SYNCHRONIZATION GROUND function	Open	Power/ground plan	OEM pin list; drawn red despite the name
U7	Pins 04/06 — sense only, or load feeds?	Open	Power sheet current rating	Measure on the live engine, or OEM pin list
U8	6-pin intake throttle — on the ECU at all?	Open	Whether a throttle driver is in scope	Trace the cable by hand — brief task 3.11; not in this diagram
U9	Catalyst temp sensor — connected where?	Open	Whether an EGT front-end is in scope	Trace on the engine
U10	SENT in the colour legend — which signal?	Open	Possible digital sensor front-end	Inspect harness; SENT is SAE J2716
U11	Engine identity — "GK550" vs Kirloskar range	Answered — Kirloskar 3R550ETA 4G1 in a KG4-25WS1, inferred, high confidence (memo 01)	Emissions tier, sensor sourcing	Engine rating plate photograph — brief task 3.1, closes outright

#	Unknown	Status	Blocks	How to close
U12	Does the 70 A ignition relay -13RB1 remove power from the ECU and fuel metering unit, or only signal the ECU?	Open	The §3 safety gate. The entire relaxation from "mandatory trip module" to "recommended" rests on this	Measure at the machine: with the ignition relay de-energised, check for battery voltage at ECU pins 21, 04, 06 and at the fuel metering unit's supply pin. Voltage present ⇒ signal only ⇒ §3 reverts. Do brief task 1.4 first — a normally-closed relay gives the same reading with the opposite meaning

U12 and U1 are now the two that matter most. U12 is the safety gate and nothing substitutes for the measurement; U1 decides whether the board mates.

U3 and U11 were the other two on that list and are now answered, both by inference from primary documents rather than by confirmation. Neither answer is fragile — but both are the kind of finding that a single photograph at the machine would settle permanently, which is why they stay on the register rather than leaving it.

U5 has moved off the critical list for a reason worth stating plainly: memo 04's family envelope means the **injector driver can be designed now, without the part number**. What the part number buys is a boost-rail target instead of a range. The thing that genuinely gates running this engine is not on this register at all — it is the injector calibration data, recorded in §9 and memo 04.

9. Out of scope

- Injection calibration: timing maps, quantity maps, rail pressure setpoint schedules. The hardware will be capable; the maps are a dyno programme, not a PCB deliverable.
- Emissions certification. **But read this before planning deployment** — research memo 06 found, in primary sources, that the regulatory position is worse than "we simply won't certify it." CPCB's RECD System & Procedure states that ECU characteristics governing injection timing, air-mass metering and emission-reduction strategy must not be altered even under CPCB's own sanctioned retrofit scheme, and excludes changes limited to engine control from that scheme entirely. General Condition 3 of GSR 804(E) states that no person shall use a genset lacking a valid Type Approval and CoP certificate. No approval pathway for a third-party ECU replacement was found.

This does not block the engineering: designing the board, building it, and bench-testing it against simulated signals are unaffected. It bears on **putting it into service on a genset that is in use**, and it compounds

the calibration risk in memo 04 — this engine meets its limits partly through calibration, so a replacement ECU can move it out of compliance without any hardware changing at all.

This is the project owners' decision to take to qualified counsel or to CPCB/ARAI directly. It is recorded here because it is material and was not previously examined, not to make the decision for them. - The GCU. We talk to the KG640C; we do not replace it. - Panel wiring, ATS, alternator metering.

10. Phase sequence

Phase	Output
0	This document
1	Research memos: engine ID, KG640C, MCU confirmation, injector and connector sourcing
2	KiCad hierarchical schematic capture
3	4-layer PCB layout, DRC, fab outputs
4	Firmware skeleton: HAL, J1939, crank sequencing, fault manager

Phase 2 cannot start on the sheets blocked by U1, U2, U5 and U8. It can start on power, CAN, MCU support, speed inputs and the relay outputs, which are fully specified above.